

There is one very important thing to understand when working with exponential moving averages, however. This is a classic source of confusion with many technical analysis indicators, i.e., the age-old question about why your exponential moving average is not the same as mine. While it may sound as if a 40-day exponential moving average only uses and only needs 40 daily closes, that's not the case. The exponential weighting starts with the previous value, and that means that it never really drops data points. It merely down-weights them into oblivion over time.

What is important to understand here is that you will get a slightly different value for a 40-day exponential moving average if you apply it on a half-year time series history for a market, than if you apply it on ten years' worth of history for the same market. I guess that what I'm trying to say here is, please refrain from sending me emails about why your EMA is a little different from mine.

Once we know that we are in a bull market, we will look for pullbacks. An easy way to quantify pullbacks is to measure the distance from a recent high, but this needs to be put into some sort of context. Clearly it wouldn't make sense to trigger a trade on a pullback of ten dollars, for instance, as a ten-dollar pullback in gold is quite different from the same in gasoline. Neither would percentage pullbacks make any sense, given the vastly varying volatility across the futures markets.

No, what we need to do is to standardise the pullback to the volatility of each market. A simplified way of measuring volatility is the average true range (ATR), as we used earlier for position sizing, so we'll keep using that. What you will find in the actual financial industry is that standard deviation, in its various forms, is used to measure volatility, but for our purposes here, the ATR is good enough.

So in this model we will use a 40-day standard deviation of price changes both for position-sizing purposes and for pullback. A pullback will be measured as the difference between the current price and the highest close in the past 20 days, divided by the standard deviation. That will tell us how many standard deviations we are away from the highest price in about a month.

This results in an analytic that can be compared across markets. You could use the same process for stocks, bonds, commodities, and other markets, as it takes the volatility into account. In this model, we are going to buy to open if we are in a bull market, as defined by the moving averages described above, and we see a pullback of three times the ATR.